Blessed G/Michael Catholic Secondary School

1st Semester Chemistry final examination for Grade 12th

Jan., 2018 E.C

Time Allowed 1:30 min'

Name Hanna Ayalew Section _____ No _____

R = 8.314J/ mol K = 0.0821 Latm/ mol K

1F = 96500C/ mol e

Molar masses : Al = 27 g/ mol , Ag = 108 g/mol , Cu = 64 g/mol , H_2 = 2g/ mol

I. Choose the best answer from the given alternatives.

B 1. A weak acid HA has a Ka = 1×10^{-5}. Which change will increase the pH of its aqueous solution the most?

A . Diluting the solution

B. Adding NaCl

C. Adding HCl

D. Lowering the temperature

B 2. Which of the following pairs will form a buffer solution?

A. HCl and NaCl

B. NH_4Cl and NH_4OH

C. NaOH and NaCl

D. H_2SO_4 and $NaSO_4$

B 3. The conjugate base of an acid is strongest when the acid has:

A. Large Ka value

B. Small Ka value

C. High concentration

D. Low molecular massive

C 4. Which statement about a buffer solution is correct?

A. Its pH does not change on adding strong acid

B. It completely neutralizes added acid

C. It resists change in pH on addition of small amounts of acid or base

D. It always has pH = 7

B 5. Which of the following salts will produce a basic solution in water?

A. NH_4Cl B. Na_2CO_3 C. NaCl D. KNO_3

B 6. Which solution has the highest pH?

A. 0.1 M HCl B. 0.1 M NaOH C. 0.1 M CH_3COOH D. 0.1 M NH_4Cl

B 7. Which of the following correctly represents the relationship between Ka and Kb of a conjugate acid–base pair at 25°C?

A. Ka + Kb B. Ka × Kb = Kw C. Ka − Kb = Kw D. Ka / Kb = Kw

B 8. At 25°C, the pH of pure water is 7. If temperature is increased, the pH of pure water will:

A. Increase B. Decrease C. Remain 7 D. Become zero

C 9. A weak acid has pKa = 4. If the ratio of salt to acid is 100, the pH of the buffer is:

A. 2 B 4 C. 6 D. 8

10. During the titration of a weak acid with a strong base, the PH at half–equivalent point is equal to

A . 7 B. pKa of the acid C. pKb of the base D. 14 – pKa

11. 20 mL of H_2SO_4 solution requires 40 mL of 0.5 M NaOH for complete neutralization.

The molarity of H_2SO_4 is:

A. 0.25 M B. 0.5 M C. 1.0 M D. 2.0 M

12. Given I. $Cu^{2+} + 2e \rightarrow Cu$, $E^0 = 0.34$ V

II. $Cu^{2+} + e \rightarrow Cu^+$ $E^0 = 0.15$ V, what it the electrode potential,

E^0, for the reaction $Cu^+ + e \rightarrow Cu$

A. 0.53 v B. 0.38 V C. 0.3 V D. 0.9 V

13. According to Faraday's second law, mass deposited is proportional to:

A. Atomic mass B. Equivalent mass C . Density D. Molarity

14. The electrode potentials of half-cell reactions

are: $Zn^{2+} + 2e^- \rightarrow Zn$ $E^0 = -0.76$ V and

$Fe^{2+} + 2e^- \rightarrow Fe$ $E^0 = 0.41$ V. The emf of the cell

$Fe^{2+} + Zn \rightarrow Zn^{2+} + Fe$ is

A. + 0.35 V B. –1.17 V C. –0.35 V D. + 1.17 v

15. The standard reduction potential of metals A,B,C and D are -0.40v, +0.34v,+0.8v and -1.67v.

The metal that can displace all others is

A . D B. C C. A D. B

16. The standard emf of a galvanic cell involving cell reaction with n = 2 is formed to be 0.295 V at 25° C. The equilibrium constant of the reaction would be

A. 1.0×10^{10} B. 2.0×10^{11} C 4.0×10^{12} D. 1.0×10^{2}

17. Which response contains all the following statements that are correct, and no others?

I. In voltaic cells the flow of electrons is spontaneous.

II. In electrolytic cells electrons flow in the external circuit (through the wire) from the anode to the cathode.

III . In voltaic cells the cathode is the positive electrode.

A. I B. I and III C. II and III D . I, II and III

18. Calculate ΔG for the reaction of the cell below under the stated conditions.

$Zn|Zn^{2+}(1.0 \times 10^{-8}\ M)||Cu^{2+}(1.0 \times 10^{-6}\ M)|Cu$

$E^0(Cu^{+2}/Cu) = 0.34$ v, $E^0(Zn^{+2}/Zn) = -0.76$ v

A . -163 kJ B. -192 kJ C. 201 kJ D . -224 kJ

19. Which of the following cells is a concentration cell?

A. $Zn|Zn^{2+}||Cu^{2+}|Cu$

B. $Cu|Cu^{2+}(0.1\ M)||Cu^{2+}(1\ M)|Cu$

C. $Ag|Ag^{+}||Cu^{2+}|Cu$

D. $Fe|Fe^{2+}||Fe^{3+}|Fe^{2}$

20. For the cell described in equation (19), EMF at 298 K is:

A. 0.118 V B. 0.059 V C. 0.0296 V D. 0 V

21. Why is standard reduction potentials not multiplied by stoichiometric coefficients?

A. Because electrons cancel C. Because Gibbs energy is additive

B. Because potential is an intensive property D. Because equilibrium constant changes

22. Which statement is FALSE for a dead battery?

A. $\Delta G = 0$ B. $E_{cell} = 0$ C. Reaction quotient $Q = K$ D. $\Delta G^\circ = 0$

23. Calculate Volume of H_2 at STP after passing 1/2 F through water:

A. 11.2 L B. 22.4 L C. 5.6 L D. 44.8 L

24. Clinker is obtained by heating:

A. Limestone + clay + silica B. Limestone + gypsum C. Clay + sand D. Limestone only

25. Which of the following types of glass is heat-resistant?

A. Soda-lime glass B. Borosilicate glass C. Lead glass D. Flint glass

26. Which of the following is a primary raw material for ceramics?

A. Clay ($Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$) B. Na_2CO_3 C. Limestone D. CaO

27. Which by-product of sugar industry is used as fermentation substrate?

A. Bagasse B. Molasses C. Lime D. Gypsum

28. Synthetic detergents are more effective in hard water because:

A. Do not form insoluble salts with Ca^{2+} or Mg^{2+} C. Contain NaOH

B. Are acidic D. Have higher density

29. Which reaction regenerates ammonia in the Solvay process?

A. $CaCO_3 \longrightarrow CaO + CO_2$

B. $NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

C. $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$

D. $NH_3 + H_2O \longrightarrow NH_4OH$

30. Which reaction represents the final step to obtain concentrated H_2SO_4?

A. $SO_3 + H_2O \rightarrow H_2SO_4$ C. $SO_2 + H_2O \rightarrow H_2SO_4$

B. $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$ D. $H_2SO_4 \rightarrow H_2O + SO_3$

31. Which of the following is NOT a pesticide?

A. DDT B. Malathion C. 2,4-D D. Urea

32. Which reaction demonstrates the dehydrating nature of H_2SO_4?

A. $Cu + H_2SO_4 \rightarrow CuSO_4$ C. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

B. $C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$ D. $SO_3 + H_2O \rightarrow H_2SO_4$

33. Which of the following pulping methods gives the highest yield?

A. Kraft pulping B. Sulfite pulping C. Mechanical pulping D. Semi-chemical pulping

B 34/ Which preservation method removes moisture by sublimation?

A. Drying B. Freeze-drying C. Smoking D. Canning

D 35. A 0.1 M solution of HA ($K_a = 1 \times 10^{-7}$). Its pH is approximately:

A. 2.0 B. 2.5 C. 3.0 D. 4

$\frac{x^2}{10^{-1}} = 10^{-7}$ $x = 10^{-4}$

A 36. In the reaction:

$HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$ (acid, base, CB, CA), The conjugate acid-base pairs are:

A. HCO_3^- / CO_3^{2-} and H_3O^+ / H_2O C . H_2O / HCO_3^- and CO_3^{2-} / H_3O^+

B. HCO_3^- / H_2O and CO_3^{2-} / H_3O^+ D. HCO_3^- / H_3O^+ only

B 37. When soap is added to oily dirt in water, cleansing occurs mainly because:

A. Soap reduces the surface tension of water permanently

B. Soap molecules form micelles that trap oil droplets

C .Soap reacts chemically with dirt to decompose it

D. Soap increases the solubility of oil in water

35/40 Very good

B 38. Which property distinguishes chemical industries from other manufacturing industries?

A. Seasonal production C. Use of manual labor

B. Transformation of substances through chemical reactions D. Production of perishable goods

C 39. Number of electrons involved in the electrodeposition of 64 g of Cu from a solution of $CuSO_4$

A. $6.02*10^{23}$ B. $3.01*10^{23}$ C. $1.204*10^{24}$ D. $6.02*10^{22}$

C 40. On the basis of information available from the reaction

$\frac{4}{3}Al + O_2 \rightarrow \frac{2}{3}Al2O_3, \Delta G = -827kJ$, The minimal emf required to carry out electrolysis of Al_2O_3 is

A. 6.42V B. 8.56V C. 2.14V D. 4.28V

Name Hanna Ayalew Section 12th C No 24.

1. B/A	9. C	17. B/D	25. B	33. C
2. B	10. B	18. D	26. A	34. B
3. B	11. B	19. B	27. A/B	35. D
4. C	12. A	20. C	28. A	36. A
5. B	13. B	21. B	29. C	37. B
6. B	14. D	22. C/D	30. B	38. B
7. B	15. A	23. C	31. D	39. C
8. B	16. D/A	24. A	32. C/B	40. C